

Demonstration of Proposed Features

Date	04 July 2026
Team ID	ramanharshitha2005@gmail.com
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Crop Recommendation (Scenario 1)	POST /api/recommend — RF+DT+SVC ensemble predicts best crop from 9 inputs with confidence % and top 10 candidates.	Implemented	Yes	Live demo: entered N=90, P=42, K=43, temp=20, humidity=82, pH=6.5, rainfall=202 → predicted rice with 94% confidence.
2	Crop Suitability Assessment (Scenario 2)	POST /api/suitability — weighted 7-metric scoring with remediation steps for the chosen crop.	Implemented	Yes	Demonstrated: selected Mango with temperature=5°C → Poor suitability, temperature remediation step shown.
3	Fertilizer Recommendation	POST /api/fertilizer — NPK gap analysis with crop-specific nutrient advice and pH note.	Implemented	Yes	Shown for Rice (nitrogen=45 — Low) → advice to increase nitrogen with split-application plan.
4	Symptom Disease Detection	POST /api/disease — RF classifier on crop+symptoms+environment returns disease + risk level.	Implemented	Yes	Demonstrated: Tomato + Late Blight symptoms + temp=24, humidity=82 → Late Blight, High risk.
5	Image Disease Detection	POST /api/image-disease — CNN or pixel-feature RF classifies uploaded leaf photo.	Implemented	Yes	Uploaded a tomato leaf image → disease classification returned with confidence score.
6	Real-time Weather Auto-fill	GET /api/weather — Open-Meteo API returns temperature, humidity, rainfall for city or GPS.	Implemented	Yes	Live demo: entered 'Nellore' → temperature, humidity, 30-day rainfall auto-filled into the recommendation form.
7	Research Analytics Dashboard (Scenario 3)	GET /api/analytics + Chart.js — bar, radar, scatter charts; filter by crop/region/season; CSV export.	Implemented	Yes	Demonstrated: filtered by Rice + South India + Kharif → charts updated; exported filtered CSV.
8	Responsive Web UI (5 pages)	index.html, recommend.html, disease.html, insights.html, contact.html with mobile hamburger menu.	Implemented	Yes	Demoed on Chrome Desktop and Chrome mobile emulator — responsive layout confirmed.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8
Overall Implementation Rate (%)	100%